Experiment-2

Aim:-

To convert the designed ER diagram into a relational schema and implement it using MySQL with primary keys, foreign keys, NOT NULL, Unique ON DELETE CASCADE, and ON DELETE SET NULL constraints.

Relational schema

```
CUSTOMER(
    Customer_ID PK,
    First_Name,
    Middle_Name,
    Last_Name,
    Email UNIQUE,
    DOB
)
CUSTOMER_PHONE(
    Customer_ID PK, FK,
    Phone PK
)
ADDRESS(
    Address_ID PK,
    Customer_ID FK,
    House_No,
    Street,
    Area,
    City,
    State,
    PIN_Code,
    Address_Type
)
SELLER(
    Seller_ID PK,
    Seller_Name,
    Email UNIQUE,
    Phone UNIQUE
)
```

CATEGORY (
    Category_ID Pk,
    Category-Name UNIQUE,
    Description
)

PRODUCT (
    Product_ID Pk,
    Product_Name,
    Price,
    Stock,
    Description,
    Seller_ID Fk,
    Category_ID Fk
)

ELECTRONICS (
    Product_ID Pk, Fk,
    Warranty_Period
)

CLOTHING (
    Product_ID Pk, Fk,
    Size,
    Fabric
)

GROCERY (
    Product_ID Pk, Fk,
    Expiry_Date,
    Weight
)

ORDERS (
    Order_ID Pk,
    Customer_ID Fk,
    Address_ID Fk,
    Order_Date,
    Status,
    Total Amount
)

ORDER_ITEM (
    Order_ID Pk, Fk,
    Product_ID Pk, Fk,
    Quantity,
    Unit_Price
)

```
PAYMENT (
    Payment_ID PK,
    Order_ID FK UNIQUE,
    Payment_Date,
    Amount,
    Payment_Method,
    Payment_Status
    Tracking_Number UNIQUE
)
```

MySQL source code

```
CREATE DATABASE Indian_Ecommerce;
USE Indian_Ecommerce;
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Middle_Name VARCHAR(50),
    LAST_Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    DOB DATE
);
CREATE TABLE CustomerPhone (
    Customer_ID INT NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    PRIMARY KEY (Customer_ID, phone),
    FOREIGN KEY (Customer_ID)
        REFERENCES (Customer (Customer_ID)
        ON DELETE CASCADE
);
CREATE TABLE Seller(
    Seller_ID INT Primary key,
    Seller_Name VARCHAR(100) NOT NULL,
    Email VARCHAR (100) NOT NULL UNIQUE,
    Phone VARCHAR(15) UNIQUE.
);
CREATE TABLE Category (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(100) NOT NULL UNIQUE,
    Description VARCHAR (255)
);
```

```
CREATE TABLE Address(
    Address_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    House_No VARCHAR(20) NOT NULL,
    Street VARCHAR(100) NOT NULL,
    Area VARCHAR(100),
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    PIN_Code VARCHAR(10) NOT NULL,
    Address_Type VARCHAR(20) NOT NULL,

    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE
);
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL,
    Description VARCHAR(255),
    Seller_ID INT NOT NULL,
    Category_ID INT NOT NULL,

    FOREIGN KEY (Seller_ID)
        REFERENCES Seller(Seller_ID)
        ON DELETE CASCADE,

    FOREIGN KEY (Category_ID)
        REFERENCES Category (Category_ID)
        ON DELETE SET NULL
);
CREATE TABLE Electronics (
    Product_ID INT PRIMARY KEY,
    Warranty-Period INT NOT NULL,
    FOREIGN KEY (Product_ID)
        REFERENCES Product(Product_ID)
        ON DELETE CASCADE
);
CREATE TABLE Clothing(
    Product_ID INT PRIMARY KEY,
    Size VARCHAR(20) NOT NULL,
    Fabric VARCHAR(50),

    FOREIGN KEY (Product_ID)
        REFERENCES Product (Product_ID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Grocery (
    Product_ID INT PRIMARY KEY,
    Expiry_Date Date NOT NULL,
    Weight DECIMAL (8,2),
    FOREIGN KEY (Product_ID)
        REFERENCES Product (Product_ID)
        ON DELETE CASCADE
);
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Customer_ID INT NOT NULL,
    Address_ID INT,
    Order_Date DATE NOT NULL,
    Status VARCHAR(30) NOT NULL,
    Total_Amount DECIMAL (10,2) NOT NULL,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (Customer_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Address_ID)
        REFERENCES Address (Address_ID)
        ON DELETE SET NULL
);
CREATE TABLE OrderItem (
    Order_ID INT NOT NULL,
    Product_ID INT NOT NULL,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL (10,2) NOT NULL,
    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE,
);
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Order_ID INT NOT NULL UNIQUE,
    Payment_Date DATE NOT NULL,
    Amount DECIMAL (10,2) NOT NULL,
    Payment_Method VARCHAR (30) NOT NULL,
    Payment_Status VARCHAR (30) NOT NULL,
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
);
```

```
CREATE TABLE Delivery (
    Delivery_ID INT Primary key,
    Order_ID INT NOT NULL UNIQUE,
    Delivery_Date DATE,
    Delivery_Status VARCHAR(30) NOT NULL,
    Tracking_Number VARCHAR(50) UNIQUE,
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
);
```

Sample data -

```
INSERT INTO Customer
Values
(1, 'Ayush', NULL, 'Singh', 'xyz@gmail.com', '2005-01-01'),
(2, 'Rahul', 'Kumar', 'Sharma', 'rahul@gmail.com', '2004-08-20');

INSERT INTO CustomerPhone
 Values
 (1, '9876543210'),
 (1, '9123456780'),
 (2, '9988776655');

INSERT INTO SELLER
VALUES
(101, 'Delhi Electronics', 'seller1@gmail.com', '9876500001'),
(102, 'Fashion Hub India', 'seller2@gmail.com', '9876500002');

INSERT INTO Category
VALUES
(201, 'Electronics', 'Electronic product'),
(202, 'Clothing', 'Clothing and fashion products'),
(203, 'Grocery', 'Daily grocer products');

INSERT INTO Address
VALUES
(301, 1, '12A', 'MT Road', 'Central Delhi',
      'Delhi', 'Delhi', '110001', 'Home'),

(302, 2, '45B', 'Park Street', 'New Delhi',
      'Delhi', 'Delhi', '110002', 'Home');

INSERT INTO PRoduct
 VALUES
 (401, 'Samsung Smartphone', 1, 31000.00, 20,
       'Android Smartphone', 101, 201),
```

```
(402, 'Cotton T-shirt', 799.00, 50,
      'Men cotton t-shirt', 102, 202),
(403, 'Basmati Rice', 999.00, 30,
      'Premium rice', 101, 203);

INSERT INTO Electronics
VALUES
(401, 24);

INSERT INTO Clothing
VALUES
(402, 'L', 'Cotton');

INSERT INTO Grocery
VALUES
(403, '2027-08-01', 5.00);

INSERT INTO Orders
VALUES
(501, 1, 301, '2026-08-18', 'Confirmed', 25798.00);

INSERT INTO OrderItem
VALUES
(501, 401, 1, 24999.00),
(501, 402, 1, 799.00);

INSERT INTO Payment
VALUES
(601, 501, '2026-08-18', 25798.00
      'UPI', 'Successful');

INSERT INTO Delivery
VALUES
(701, 501, NULL, 'Processing', 'DEL123456');
```

Demonstrating ON DELETE CASCADE

```
DELETE FROM Orders
WHERE Order_ID = 501;
```

Verify -

```
SELECT * FROM OrderItem;
SELECT * FROM Payment;
SELECT * FROM Delivery;
```

Demonstrating ON DELETE SET NULL

```
DELETE FROM Address
WHERE Address_ID = 301;
```

The order is not deleted. Instead

```
SELECT * FROM Orders
WHERE Order_ID = 501;
```

Use this correct definition -

```
Category_ID INT,
FOREIGN KEY (Category_ID)
    REFERENCES Category (Category_ID)
    ON DELETE SET NULL
```

Instead of:

```
Category_ID INT NOT NULL
```

Output section -

Query OK, 0 rows affected